



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

Lecture 1: Course Introduction

SC6125 – Blockchain in Finance



Dr. Ernie Teo

LinkedIn

Course Introduction

- This course provides an overview of the applications of blockchain in finance
- It is designed for students who are interested in the applications of blockchain in finance
- The course includes innovative applications of finance in the areas of DeFi as well as blockchain use cases of traditional financial institutions
- You will learn to analyze and critique different blockchain projects in finance as well as to design applications for finance using blockchain

By the end of this course, you should be able to:

1. Describe applications of blockchain for finance
2. Understand and compare the different finance applications for blockchain (decentralized and centralized)
3. Explain the technologies underlying finance and blockchain use cases
4. Design DeFi applications and enterprise applications for finance



Syllabus

Week	Topic
1 13 Apr	Course Introduction Fintech 4.0 and Web3
2 20 Apr	DeFi 1 Tokens and Fundraising, NFTs, Stablecoins
3 27 Apr	DeFi 2 DeFi building blocks: Lending, Exchanges, Savings, Insurance
4 4 May	Quiz (covers Weeks 1 to 3)
5 11 May	Guest Lecture by Schroders Financial Institutions and Blockchain – Government projects from Singapore and other countries
6 18 May	Group Project Presentation



Assessment

Component	Weighting	Team/Individual
Quiz	40%	Individual
Class Participation	20%	Individual
Group project (Presentation and Written Summary)	40%	Team
Total	100%	

Quiz – 40%

- E-quiz to be done in class (in Week 4)
- Covers Week 1 to 3
- 90 minutes
- Format:
 - 20 Multiple Choice Questions (10%)
 - 2 Essay/Short Answer Questions (30%)

Class Participation – 20%

- Wooclap – 10%
 - Participate in Wooclap activities
 - Answer questions correctly
- Guest Lecture Reflection – 10%
 - To be done on NTULearn after the lecture
- Bonus marks
 - Ask questions in class



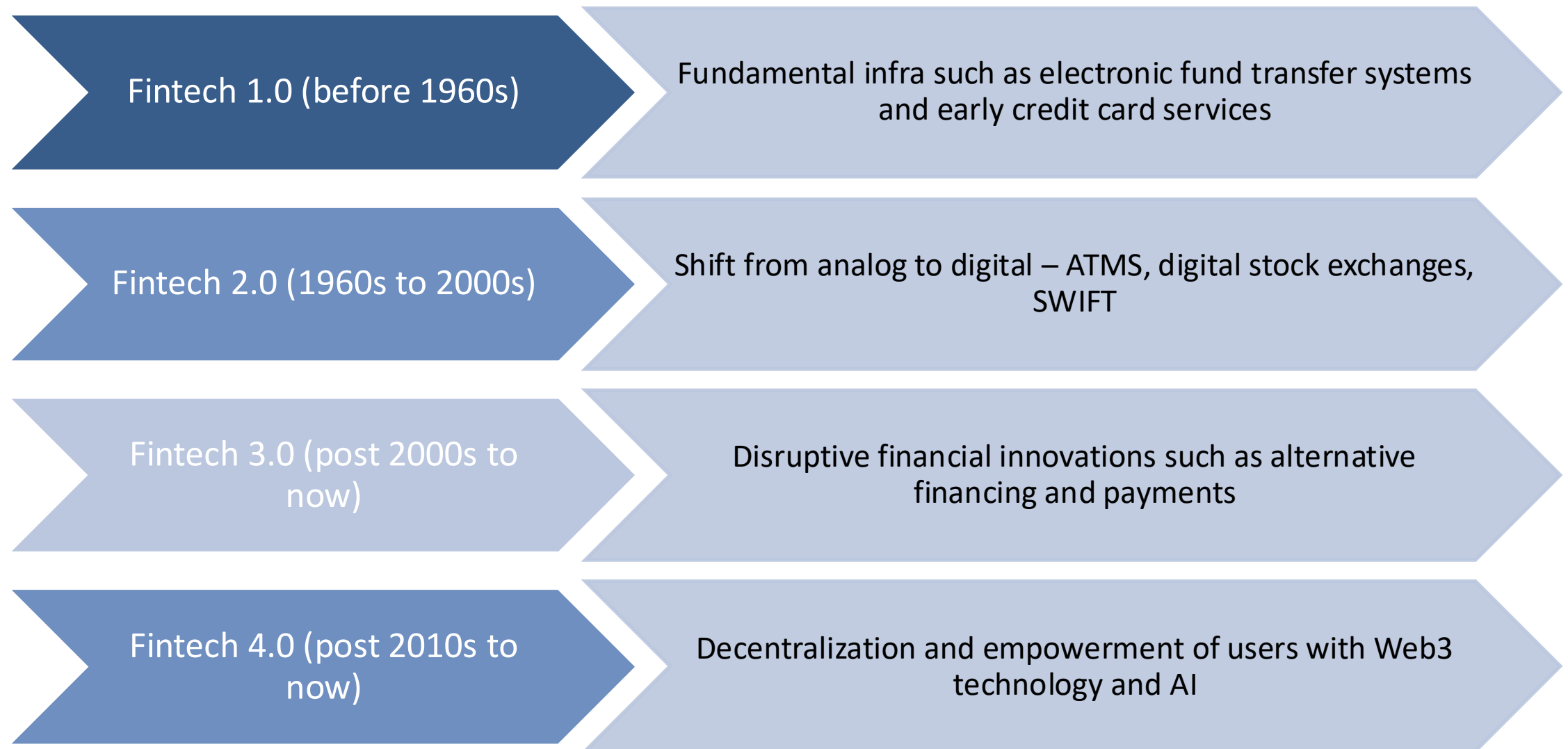
Fintech

“Computer programs and other technology used to support or enable banking and financial services” – Oxford dictionary

Do you agree?



Versions of Fintech



What should FinTech be?

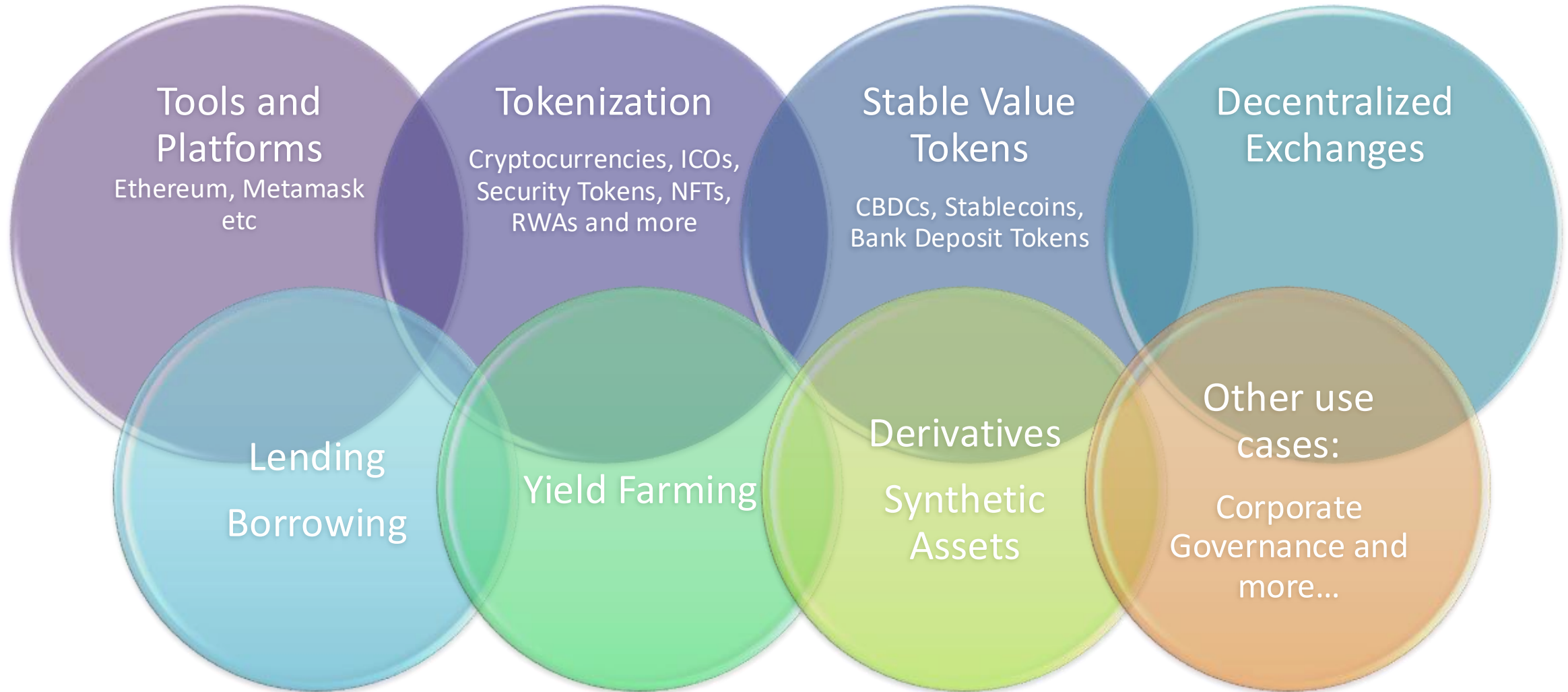
FinTech should be **innovative** in offering financial services through **technology and user experience**

FinTech is no longer just disruptive

Fintech 1.0 to 4.0: Across Finance

	Fintech 1.0 – 2.0	Fintech 3.0	Fintech 4.0
Payments	Credit/Debit Cards Bank Cards	QR Codes NFC Online Payment Processors APIs as a Service	Cryptocurrencies Stable-coins CBDCs
Remittance	Centralized Messaging Networks and Money Transfer Services	Peer to Peer FX Networks Telco Credits and SMS	Crypto Transfers
Financing	Bank Financing Credit Unions	P2P Borrowing Crowdfunding Buy Now Pay Later	Crypto Borrowing
Savings and Investments	Bank Deposits Securities Commodities	Micro-investments Robo-advisors Alt Lending	Staking Digital Assets and Tokens Decentralized Exchanges Other DeFi Products
Insurance	Traditional Insurance Manual Underwriting	Parametric Insurance Data Driven Underwriting	Smart Contract Driven (Underwriting and Claims) P2P Mutual Insurance
Compliance and Regulations	Manual KYC and AML Paper Docs and Wet Sign	AI Driven KYC and AML Digital Business Process	DID, Acc Abstractions Decentralized Blockchain Evidence and Process

Web3 and Decentralized Finance



DeFi vs Traditional Finance

DeFi	Traditional Finance
Smart Contracts and Decentralized Networks	Intermediaries: Banks, Brokers, Regulators
Bring your own wallet and identity	Centralized Permissions and Sign Ups required
Transparent, Verifiable and Immutable	Lacks Transparency and Accountability
Globally available	Limited access due to intermediaries and regulations
Low Fees and Fast Transactions	High Fees and Slow Transactions
Rapid Competition and Innovation	Limited Competition

DeFi vs Traditional Finance

DeFi	Traditional Finance
Smart Contracts and Decentralized Networks	Intermediaries: Banks, Brokers, Regulators
Bring your own wallet and identity	Centralized Permissions and Sign Ups required
Transparent, Verifiable and Immutable	Lacks Transparency and Accountability
Globally available	Limited access due to intermediaries and regulations
Low Fees and Fast Transactions	High Fees and Slow Transactions
Rapid Competition and Innovation	Limited Competition

The lines are now blurring

Web3 in Singapore

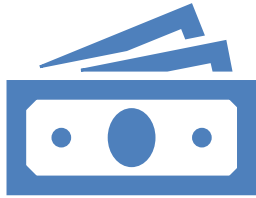
"The Future of Money, Finance and the Internet"

**Speech by Mr Ravi Menon, Managing Director,
Monetary Authority of Singapore, at Singapore
FinTech Festival on 9 November 2021**

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Juno, Plutus, Mercury



Money



Finance



Commerce and
Communications

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>

Juno: Money

- Commercial banks create money
- Central banks preserve its value
- Most money in Singapore are created by commercial banks and in digital form
- 92% of money supply in Singapore is in bank deposits and are used for electronic payments
- Commercial banks in turn place reserves with the central bank to settle interbank transactions
- Confidence in money is anchored by the central bank

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>

Juno: Money

- Three new developments in money that are challenging the status quo
 - Cryptocurrencies
 - Stablecoins
 - Central Bank Digital Currencies

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Juno: Money - Cryptocurrencies

- MAS prefers the term crypto tokens
- Tokens used for payments are subject to MAS licensing and supervision
- Crypto tokens for investment are frowned upon as there are risks of suffering significant losses
- MAS also acknowledges that blockchain and crypto tokens can bring many potential benefits
 - Such as to facilitate cross border payments and trade finance
- To be regarded as money crypto tokens need to be more stable in value

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>

Juno: Money - Stablecoins

- Combines credibility of fiat money and advantages of blockchain
- Stablecoins are finding acceptance outside of the crypto ecosystem
 - Visa and Mastercard are allowing settlement using USD Coin
- Stablecoins can potentially pose financial stability risks
 - Runs on the issuer on the stablecoin
- MAS needs to regulate appropriately and tighten if necessary

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Juno: Money - CBDC

- A direct liability of and payment instrument issued by central bank
- Wholesale – restricted to use within banking system
 - Can transform cross-border payments
- Retail – issued to the public and are digital equivalent of cash
 - 6 out of 10 central banks are experimenting with retail CBDCs

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Juno: Money – Retail CBDC

- Benefits of central bank issued digital money
 - Widely accepted
 - Bears the authority of state
 - Ultimate risk-free asset
 - Means of final settlement
- More efficient and inclusive payment ecosystem
 - Startups can integrate and not have to build their own e-money
 - Financial inclusion
- Mitigate against stablecoins or foreign CBDCs
 - May displace the use of Singapore dollar in domestic transactions

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>

Juno: Money – Retail CBDC

- Retail CBDCs can potentially pose significant risks to monetary and financial stability
 - People can switch from bank deposits to risk-free central money quickly
 - Reduce capacity of banks to make loans
 - Design with safeguards such as stock and flow caps
- Retail CBDC issuance is not urgent for Singapore
 - Need for digital cash is moot at this point in time
 - No compelling financial inclusion benefits
 - Risk of foreign currency substitution low
- Project Orchid – to build technology infrastructures and competencies for the future

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>

Plutus: Finance

- Two developments shaping the future of finance
 - Real time cross border payments
 - Collaborative data platforms

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Plutus: Finance – Cross Border Payments

- Project Ubin to make cross-border payments cheaper, faster and more efficient
 - Has inspired Patior – a blockchain based interbank clearing and settlement network
 - Jointly established by DBS Bank, JP Morgan and Temasek
 - Enables banks to settle cross border in different currencies in real time using commercial bank digital money or wholesale CBDCs
- Project Dunbar – a blueprint for multi-currency settlement platform that operates across countries using wholesale CBDCs
 - Partnership among Bank for International Settlement, MAS, Reserve Bank of Australia, Bank Negara Malaysia and South Africa Reserve Bank
 - In March 2022, it was announced that Project Dunbar has successfully completed the prototypes for a common platform enabling international settlements using m-CBDCs

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Plutus: Finance – Cross Border Payments

- PayNow, Singapore's real time retail payment system
 - Is building direct linkages with other countries' payment solutions such as Thailand (PromptPay), Malaysia (DuitNow) and India (Unified Payment Interface)
 - Bilateral payment links one at a time is time consuming
 - A multilateral solution is needed
 - Project Nexus with BIS aims to create a common blueprint for how countries can fully integrate their real time payment systems into a single cross border network

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Plutus: Finance – Collaborative Data Platforms

- Project Greenprint - a technology and data platform to support the green finance system
 - Common disclosure portal to make ESG disclosures
 - Data orchestrator to aggregate ESG data
 - ESG registry – to record and maintain ESG certifications on distributed ledger
 - Greenprint marketplace to connect green technology providers with investors and corporates

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Plutus: Finance – Collaborative Data Platforms

- COSMIC – a platform for financial institutions to collaborate using data analytics to combat the risk of money laundering
- Will enable seamless data exchange with financial institutions' own data analytics systems so that they can identify and disrupt criminal networks sprawling across multiple financial institutions
- MAS has been co-creating COSMIC with six major banks in Singapore and will go live in 2023

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Mercury: Commerce and Collaborations – Web3

- Web 1.0 is the “readable” Internet, centered on access to information and where most users were consumers of content. It is about web pages, emails, and chatrooms.
- Web 2.0 is the “interactive” Internet, with rich exchanges of information and user-generated content and collaboration. Key features are blogging, tagging, and social media.
- Web 3.0 is the “personal” Internet, empowering end-users through applications that allow the decentralized sharing of information. The key enablers of this new paradigm are smart contracts and tokenized assets.

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>

Mercury: Commerce and Collaborations – Web3

- Smart Contracts and Tokenization
- Smart contracts are already being used to enhance the market infrastructure for financial assets in Singapore for digital bonds issuance
 - Smaller denominations
 - Shorter issuance settlement
 - Automated coupon and redemption payments
- Project Guardian focuses on asset tokenization and its various use cases

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Mercury: Commerce and Collaborations – Web3

- DeFi – Decentralized Finance
 - No financial intermediaries
 - Potential to yield significant economic and social benefits
 - Reduce cost of finance
 - More competition and inclusion
 - Has its risks and vulnerabilities
 - Have yet to meet high standards of governance, security and resilience
 - Who is responsible?
 - Regulatory frameworks needed for DeFi to become a reality
 - More challenging as governance and control is dispersed across the blockchain

<https://www.mas.gov.sg/news/speeches/2021/the-future-of-money-finance-and-the-internet>



Web3 or Web 3.0?

- Since 2021 when this speech was given the definitions have been further refined
- Web 3.0, focuses on efficiency and intelligence by reusing and linking data across websites
- Web3, puts a strong emphasis on security and empowerment by returning control of data and identity to users
 - And is based on blockchain technology
 - Mr Ravi Menon was referring to Web3 in his speech

What do you think of this speech?

- What do you think of MAS's views in 2021?



What do you think of this speech?

- If you were to make this speech as MAS in 2026, what would you recommend?





DeFi in Singapore

- Under the Payment Services Act 2019 (PS Act)
 - Guidelines are laid out for a business that is providing a digital payment token (DPT) service
 - Protect consumers from the high risks
 - Mitigate money laundering and terrorism financing risks
- DPT service providers are required to
 - Not promote their services to the public in Singapore
 - Conduct customer due diligence and transaction monitoring
 - Implement safeguards for customer assets
 - Disclose fees and risks to customers
- 37 DPT providers have been licensed under the PS Act as of Apr 2026



DeFi in Singapore

- Aug 2023: “Stablecoin Issuance Service” announced as a new payment service regulated under the Payment Services Act
- Single Currency Stablecoins (SCS) that are under the SCS legal framework will be labelled as “MAS-regulated stablecoins”
- Single Currency Stablecoin issuers can now apply to be regulated under MAS
- StraitsX and Paxos have been given in-principle approvals to issue stablecoins in Singapore



DeFi in Singapore

- Project Guardian
 - To test the feasibility of applications in asset tokenisation and decentralised finance
 - Started in June 2022 and is still ongoing
 - To test the potential of asset tokenisation across more financial asset classes
 - Involves DBS Bank, Standard Chartered Bank, UOB, SGX and also crypto industry players



DeFi in Singapore

- Project Orchid
 - Launched in November 2021
 - Phase 1, January 2022, looked at a purpose-bound digital Singapore dollar (SGD)
 - A multi-year project pertaining to a retail CBDC system for Singapore
 - Aims to build the technology, infrastructure and technical competencies that are necessary to issue a digital SGD
 - Explore the potential benefits and challenges of CBDC for Singaporeans and the economy

MAS Announces Plans to Support Commercialisation of Asset Tokenisation

- On 4th Nov 2024, MAS announced the following plans
 - Forming commercial networks to deepen liquidity of tokenised assets
 - Developing an ecosystem of market infrastructures
 - Fostering industry frameworks for tokenised asset implementation
 - Enabling access to common settlement facility for tokenised assets
- Mr Leong Sing Chiong, Deputy Managing Director (Markets and Development) of MAS
 - “MAS has seen strong interest in asset tokenisation in recent years, notably in fixed income, FX, and asset management. We are encouraged by the keen participation from financial institutions and fellow policymakers to co-create industry standards and risk management frameworks to facilitate commercial deployment of tokenised capital markets products, and scale tokenised markets on an industry wide basis.”

<https://www.mas.gov.sg/news/media-releases/2024/mas-announces-plans-to-support-commercialisation-of-asset-tokenisation>



MAS Announces Plans to Support Commercialisation of Asset Tokenisation

1. Deepening liquidity of tokenised assets through formation of commercial networks

- Project Guardian, convened over 40 financial institutions
- More than 15 industry trials have been conducted in six currencies across multiple financial products
- MAS is facilitating commercialisation (of these projects) to take place in a coordinated, networked manner
- It aims to connect a broader set of participants' products and services across multiple currencies and assets
 - Leading to improvements in capital raising, secondary trading, asset servicing and settlement of tokenised assets
 - This will deepen liquidity across primary and secondary markets for tokenised asset transactions
- Citi, HSBC, Schroders, Standard Chartered and UOB have formed the Guardian Wholesale Network industry group, with the intent of establishing a multi-member network to commercialise their respective asset tokenisation trials and scale usage

<https://www.mas.gov.sg/news/media-releases/2024/mas-announces-plans-to-support-commercialisation-of-asset-tokenisation>



MAS Announces Plans to Support Commercialisation of Asset Tokenisation

2. Developing ecosystem of market infrastructures to facilitate seamless cross border transactions

- MAS launched the Global Layer One (GL1) initiative in 2023 to foster the development of foundational digital infrastructures, upon which commercial networks could be deployed
 - BNY, Citi, J.P. Morgan, MUFG and Societe Generale-FORGE have been leading efforts to define the business, governance, risk, legal and technology requirements of the GL1 Platform
- Going forward, GL1 will undertake the following additional activities:
 - Control Principles – Alignment on governance, risk management controls and settlement arrangement conventions for cross border transactions. This provides clarity on roles, responsibilities and controls needed to safeguard market integrity and financial stability.
 - Specifications – Development of specifications for market infrastructures and asset lifecycle. This encourages interoperability between diverse systems.
 - Compliance by Design – Creation and provision of templates including programmable compliance checks to build an ecosystem of compatible service providers. This accelerates onboarding for new participants
- To support these developments, MAS is pleased to announce the addition of new industry participants, including Euroclear and HSBC
- GL1 will also set-up a new market infrastructure working group, comprising global financial market infrastructure providers, that will focus on digital asset securities control principles

<https://www.mas.gov.sg/news/media-releases/2024/mas-announces-plans-to-support-commercialisation-of-asset-tokenisation>

MAS Announces Plans to Support Commercialisation of Asset Tokenisation

3. Industry Frameworks for Implementation of Tokenisation

- To facilitate broad based acceptance and implementation of tokenised assets by financial institutions, two industry frameworks were developed by Project Guardian industry group
- Guardian Fixed Income Framework (GFIF)
 - This provides an industry guide to implementing tokenisation in Debt Capital Markets, strengthen industry capabilities and catalyse adoption of tokenised fixed income solutions
- Guardian Funds Framework (GFF)
 - GFF provides a set of recommendations for industry best practices for tokenised funds. This includes the Guardian Composable Token Taxonomy to facilitate development of tokenised investment vehicles comprising multiple assets, simplifying the process of incorporating new tokenised funds, and help achieve efficiencies in fund settlement.

<https://www.mas.gov.sg/news/media-releases/2024/mas-announces-plans-to-support-commercialisation-of-asset-tokenisation>

MAS Announces Plans to Support Commercialisation of Asset Tokenisation

4. Access to Common Settlement Facility for Tokenised Assets

- Common settlement assets are instruments which are mutually agreed upon by transacting parties to execute financial transactions
- MAS is facilitating financial institutions' access to common settlement assets including S\$ wholesale CBDC for market testing purposes
- The initial test network (SGD Testnet) will offer three key features:
 - Settlement facility – Issuance, transfer and redemption of S\$ wholesale central bank digital currency (CBDC), with potential extensions to other forms of central bank and commercial bank liabilities.
 - Programmability – Automated and conditional triggers for tokenised transactions, including the use of Purpose Bound Money .
 - Interoperability – Facilitate linkages with existing financial market infrastructures.
- The first set of participating financial institutions includes DBS, OCBC, Standard Chartered and UOB
- Participating use cases include payments and securities settlement

<https://www.mas.gov.sg/news/media-releases/2024/mas-announces-plans-to-support-commercialisation-of-asset-tokenisation>



OCBC Integrates Blockchain Into LTA to Create Greater Transparency in Contractor Payments

by The Fintech Times · November 9, 2024

- Mobilisation advance payment helps the main contractors to defray heavy upfront capital outlays at the start of their construction project
- With OCBC's blockchain-based conditional payments solution, the mobilisation advance payments are automatically disbursed to the intended contractor once the smart contracts verify that the conditions are met
- The solution allows LTA (Land Transport Authority) to have full transparency over the use of the mobilisation advance, and integrating it within OCBC's business banking platform provides a real-time overview of the transactions and bank balances.
- The pilot involves three main contractors for the Cross Island Line projects. To date, over S\$22million has been disbursed to the main contractors and sub-contractors with this solution.
- Having actively participated since 2022 in Monetary Authority of Singapore's (MAS) multi-year, multi-phase Project Orchid to develop the technology infrastructure and technical competencies necessary for a digital Singapore dollar, OCBC has concurrently developed its own capabilities in blockchain-based conditional payments.

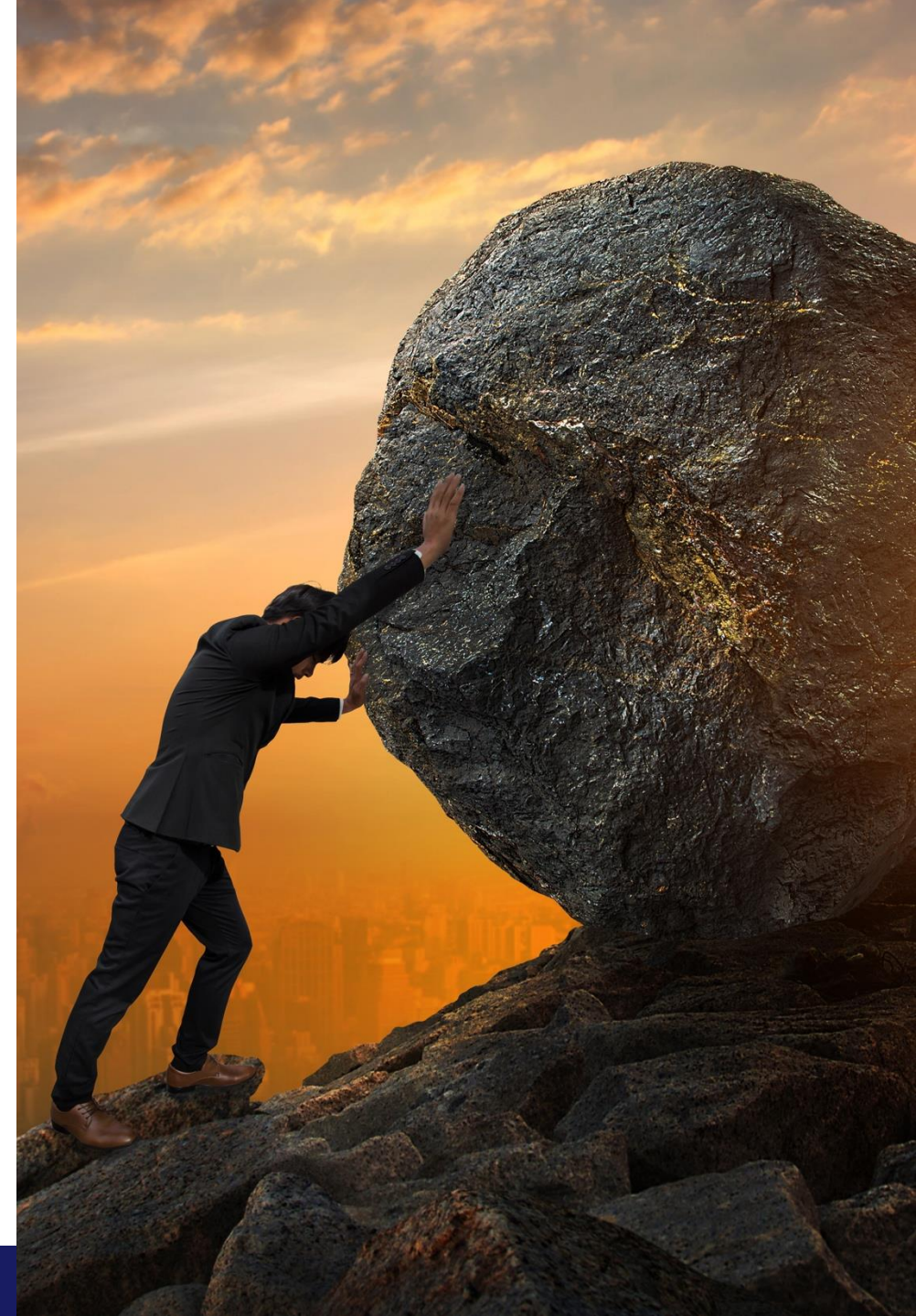
<https://thefintechtimes.com/ocbc-integrates-blockchain-into-lta-to-create-greater-transparency-in-contractor-payments/>



Changes in DeFi

The DeFi industry is an experiment in progress

- Technologists
 - Are still trying to improving the technology
 - Strengthening of security, privacy, scalability
- Regulators
 - Have moved from observing to acting
 - New regulations are being rolled out
- Startups
 - Continuing to innovate with new business models
 - Starting to get licensed and regulated
- Traditional Enterprise
 - Figuring out how to use blockchain in the right places
 - Starting to believe in the benefit of decentralization



DeFi vs Traditional Finance

DeFi	Traditional Finance
Smart Contracts and Decentralized Networks	Intermediaries: Banks, Brokers, Regulators
Bring your own wallet and identity	Centralized Permissions and Sign Ups required
Transparent, Verifiable and Immutable	Lacks Transparency and Accountability
Globally available	Limited access due to intermediaries and regulations
Low Fees and Fast Transactions	High Fees and Slow Transactions
Rapid Competition and Innovation	Limited Competition

The lines are now blurring

Blockchain has become part of finance

- Decentralization and Tokenization has driven innovations in new financial products
 - Decentralized Finance has gained traction over time
 - With a global market cap of between 80-90 billion in Nov 2024
- Centralized Finance has also been integrating with Web3 in different ways and become a part of the ecosystem
 - Custodian Services
 - ETFs
 - Use in more traditional banking – e.g. OCBC and LTA
- The lines between the two worlds are blurring as blockchain technologies mature and use cases develop further

In SC6125

- We will
 - Explore the changing role of blockchain in the role of finance
 - From tokenization to DeFi
 - To use cases of traditional financial institutions
 - And the world in between
- Through the group assignment, we will attempt to
 - Address real world issues in finance with blockchain based solutions



Short Quiz



Group Assignment 40%

- **Designing a Blockchain Enabled Financial Business Model**
- Objective
 - Design and evaluate a blockchain based financial business model that addresses a real financial problem, demonstrating both technical understanding and financial judgment
 - Projects will be assessed primarily on the quality of financial problem framing and economic logic, not the sophistication of the blockchain technology alone
- Scope
 - Your solution must involve:
 - Blockchain technology
 - A financial use case (e.g. payments, remittance, loans, investments, corporate finance, capital markets, asset management, trade finance, insurance, treasury)
- To be done in groups of 5, please register your group in NTULearn by today (13th April)



Group Assignment – 40%

- Stage 1:
 - Choose 3 areas of finance and suggest where blockchain can help
 - Week 2 – post in padlet
- Stage 2:
 - Finalize one area of finance
 - Between Week 2-3 – confirm on padlet
- Stage 3:
 - Brainstorm 3 blockchain ideas for the chosen area of finance
 - Week 3 – post on padlet
- Stage 4:
 - Finalize solution
 - Week 4 – confirm on padlet
- Stage 5:
 - Presentation and Written Summary
 - Week 6



Group Assignment 40%

Required Components

- 1. Problem Definition (Finance First)
 - Clearly define the financial problem
 - Identify users and stakeholders
 - Explain why existing solutions are inadequate
- 2. System & Architecture Design
 - High level system architecture
 - On chain vs off chain components
 - Participant roles

Group Assignment 40%

Required Components

- 3. Smart Contract / dApp Design (Mandatory)
 - Choose ONE of the following:
 - Option A — Smart Contract Logic
 - Pseudocode or Solidity style structure
 - Core functions and conditions
 - Option B — Transaction & Contract Flow
 - Sequence diagram
 - States, triggers, and outcomes
 - No requirement for deployable or fully functional code. Focus on:
 - Business logic
 - State transitions
 - Roles and trust assumptions
 - Financial consequences

Group Assignment 40%

Required Components

- 4. Digital Asset Comparison
 - Clearly justify choice of:
 - Stablecoin, CBDC, deposit token, or other digital asset
 - Compare alternatives and trade offs
 - Discuss potential challenges such as stability and issuer risks
- 5. Evaluation & Valuation
 - Benefits and limitations
 - Adoption and regulatory challenges
 - Business viability / scalability

Group Assignment 40%

- Deliverables: Presentation + Written Summary
- Presentation Duration: 20-25 minutes + Q&A
- Suggested flow:
 - Financial problem & stakeholders
 - Proposed blockchain-enabled solution
 - System and smart contract logic (high-level)
 - Digital asset choice
 - Key risks and limitations
- Submissions
 - Slides (PDF)
 - 2–3 page written summary
 - This is not a full report
 - Summarize the key points of your presentations in words



Group Assignment: Marking Criteria

Financial Problem & Context - 15%

- Financial problem is clearly defined, economically meaningful, and well-motivated
- Stakeholders are explicitly identified with clear incentives
- Strong evidence that existing solutions are inadequate

System & Architecture Design – 15%

- Clear, coherent architecture showing strong understanding of on-chain vs off-chain roles, participants, trust boundaries, and data flows
- Design choices are well-justified

Smart Contract / dApp Design – 20%

- Smart contract logic or transaction flow is clear, accurate, and well-aligned with the financial use case
- Demonstrates strong understanding of states, triggers, constraints, and risks

Group Assignment: Marking Criteria

Digital Asset Comparison – 15%

- Digital asset choice is clearly justified with thoughtful comparison of alternatives and explicit trade-offs (technical, financial, and regulatory)

Evaluation & Business Viability – 15%

- Balanced evaluation of benefits and limitations
- Demonstrates realistic understanding of adoption, regulation, scalability, and business viability

Group Assignment: Marking Criteria

Presentation Quality – 10%

- Presentation was engaging and pitched appropriately for intended target audience
- An ideal volume and speed were used for the presentation
- Visuals and citations used appropriately

Written Summary – 10%

- Content is well-structured, logical, and flow is natural
- Poster is self-explanatory and sufficient, without excessive unnecessary detail
- Good use of visuals to enhance content and/or comprehensibility
- All necessary content and visuals are cited

Group Assignment: Reference Example

(Not to Be Replicated)

- Reference Example: On-Chain Settlement Using Stablecoins for Cross-Border Payments
 - This is based on how some fintechs are already using stablecoins
 - Your project should attempt to improve on something existing instead of replicating an existing solution
- This example is provided to show the level of clarity, structure, and financial reasoning expected
 - You should not reuse this use case
 - Projects that are materially similar may be penalised
- Your project should address a different financial problem, user group, or market segment
- You should focus on how you solve the problem, not what technology you use



Group Assignment: Reference Example

(Not to Be Replicated)

Financial Problem (Finance First)

- Problem context - Cross-border payments between financial institutions are:
 - Slow (T+1 to T+3 settlement)
 - Expensive (multiple intermediaries, FX spreads)
 - Operationally complex (correspondent banking, nostro/vostro accounts)
- Who is affected
 - SMEs making international payments
 - Fintechs and payment service providers
 - Banks managing trapped liquidity across currencies
- Why existing solutions are inadequate
 - SWIFT is a messaging system, not a settlement layer
 - Pre-funding requirements tie up capital
 - Limited transparency on fees and settlement status
- Tip
 - This part should focus on the problem (without blockchain)
 - Adequate background research is important



Group Assignment: Reference Example

(Not to Be Replicated)

Proposed Blockchain-Enabled Solution

- Use a regulated, fiat-backed stablecoin as a settlement asset between participating payment institutions
- Blockchain can achieve
 - Near-instant settlement
 - Reduced intermediaries
 - Programmable settlement logic
- What blockchain does NOT replace
 - KYC/AML
 - FX conversion (still handled off-chain)
 - Customer-facing banking relationships
- Tip
 - This is just a high level example, your project should be more detailed
 - Focus on what blockchain can do and be realistic on what it cannot do

Group Assignment: Reference Example

(Not to Be Replicated)

System & Architecture Design

- Participants
 - Sending payment institution
 - Receiving payment institution
 - Stablecoin issuer
 - Blockchain network (permissioned or public)
- On-chain
 - Stablecoin transfers
 - Settlement finality
 - Smart contract-based escrow
- Off-chain
 - Customer onboarding and compliance
 - FX rate determination
 - Fiat in/out via bank accounts
- Tip
 - This is just a high level example, your project should be more detailed
 - Determine what can happen on chain and what needs to be done off chain



Group Assignment: Reference Example

(Not to Be Replicated)

Smart Contract / Transaction Flow

- Example transaction flow
 - Sending institution locks stablecoins in a smart contract
 - Contract verifies recipient institution address and amount
 - Settlement occurs atomically on-chain
 - Receiving institution credits end customer off-chain
- What the smart contract enforces
 - Who can send/receive
 - Settlement conditions
 - Optional timeouts or rollback logic
- Tip
 - Focus on logic and states, not code syntax
 - Consider unhappy paths
 - Consider when humans come in (or not)



Group Assignment: Reference Example

(Not to Be Replicated)

Digital Asset Choice

- Chosen asset: Fiat-backed stablecoin
- Alternatives considered
 - CBDC (limited availability, policy constraints)
 - Tokenised bank deposits (interoperability issues)
 - Traditional correspondent banking (status quo inefficiencies)
- Trade-offs
 - Credit risk of issuer
 - Regulatory oversight vs decentralisation
 - Public vs permissioned blockchain choice
- Tip
 - You should be choosing the digital asset based on the use case not the technology



Group Assignment: Reference Example

(Not to Be Replicated)

Evaluation & Limitations

- Benefits
 - Faster settlement
 - Reduced operational costs
 - Improved transparency
- Limitations
 - Regulatory acceptance varies by jurisdiction
 - Dependence on stablecoin issuer
 - FX still largely off-chain
- Business viability
 - Most suitable for B2B and institutional flows
 - Less suitable for retail remittance in some markets
- Tip
 - Discuss how limitations can be addressed but no need to solve it here



Group Assignment – 40%

- You can form your groups now
 - Contact me once you have formed the group to get the link to your private padlet
- From next week
 - Stage 1:
 - Choose 3 areas of finance and suggest where blockchain can help
 - Week 2 – post in padlet before class
 - Stage 2:
 - Finalize one area of finance
 - Between Week 2-3 – confirm on padlet